THE UNIVERSITY OF
MELBOURNE

COMP90050
Advanced Database Systems

Semester 1, 2026

Lecturer: Farhana Choudhury (PhD)

Live lecture – Week 1

Today's live lecture

- Subject introduction
- Logistic information
- Discussion on the topics of the pre-recorded lectures
- Additional contents

What this subject is all about

A **Database Management System (DBMS)** is a software system designed to store, manage, and facilitate access to databases.

Essentials to achieve correct behaviour and the best possible performance

- Knowledge of how DB systems work
- Mechanisms used by current systems to provide useful features
- Advanced topics

A database system should provide

- Ability to retrieve and process the data **effectively and efficiently**
- **Secure** and **reliable** storage of data

Core Concepts of Database management system

Database performance metrics

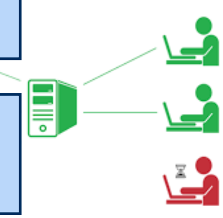
- Efficiency/speed
- Effectiveness
- Security & Reliability

Efficiency

- Hardware
  - Disks and I/O bandwidth
  - Main memory
  - Type of architecture
- Software/ DB tuning
  - Types of DB
  - Indexing
  - Query optimisation

Effectiveness

- Concurrent users: Users reading and writing over the same data
- Transactions: Required tasks are all done together



Reliability

- Crash recovery
- Fault tolerance
- Data duplication

Logistic Information

All the key information are available on Canvas:

- You need to check it a few times a week for being up to date on discussions and announcements
- It has also other basic info such as contact info for us
- Find project specifications and other assessments
- Where submissions are made
- Where you can find lecture capture and other materials

Logistic Information

Lectures: A combination of pre-recorded and live sessions

(Flipped classrooms)

- One hour lecture on Thursdays 3-4pm on-campus + will be later available in lecture capture

Tutorials: one 1-hour tutorial per week

- **Starts from Week 2**
- Tutorials are conducted on-campus

Who we are

Lecturer and subject coordinator

Dr. Farhana Choudhury

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Head tutor

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BUT Ed DISCUSSION and ANNOUNCEMENTS first!

Assessments

- Online quizzes – 5 quizzes worth 2% each (Week 3, 5, 7, 9, 11)

  -Will be open on Tuesday 3pm, closes on Thursday 3pm (right before the live lecture)

  -20 minutes strict time to complete once opened

- Project – survey report and oral presentation on a database research topic (40%) as a **group of 4** students (presentation 15% + Report 25%)

- Final examination (50%)

Project

Step 1: Form a group of **4 students** by the end of Week 3.

Step 2: Pick a topic of your interest from a list of candidate topics provided, by the end of Week 3.

Canvas has details about the projects already: **Read** and **Start** with the steps above.

Presentation: Around Week 11 (schedule will be uploaded)

Report submission due data: 25 May 2026, 11:59pm

Discussion on the topics of the pre-recorded lecture and additional contents

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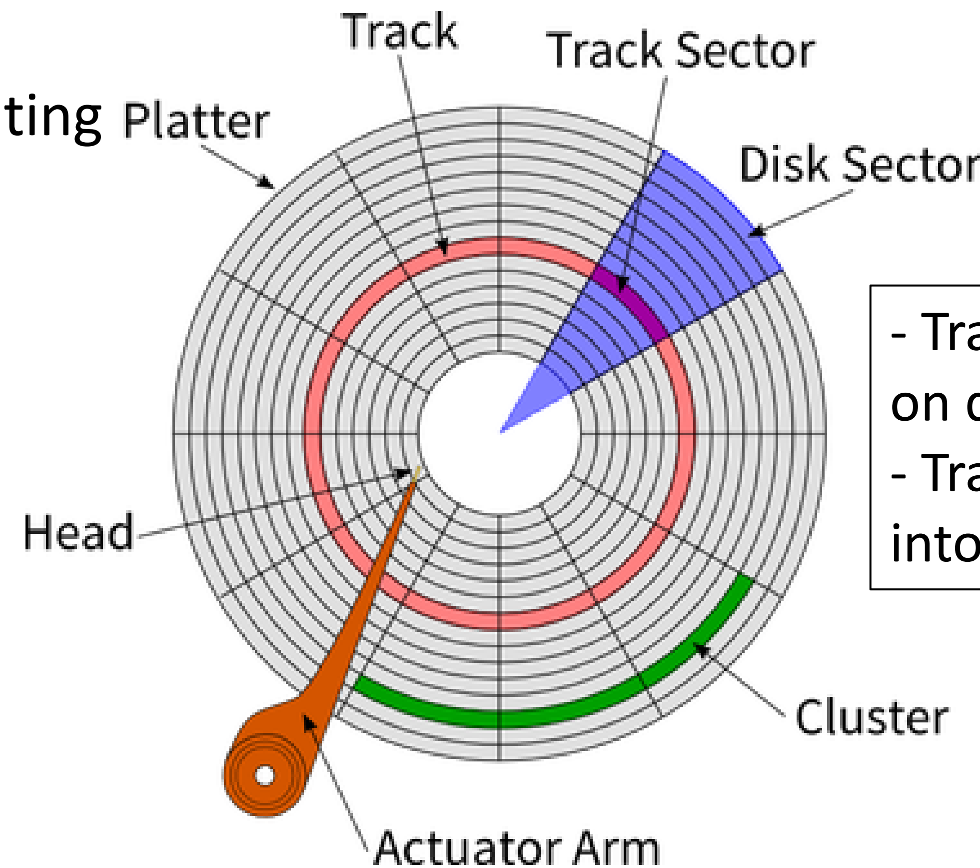
Basic Hardware of a classical disk

Rapidly rotating



- Tracks: circular path on disk surface
- Tracks are subdivided into disk sectors

with magnetic head, which reads and writes data to the platter surfaces

SSD (Solid-State Drive/Solid-State Disk)

- **No moving parts** like Hard Disk Drive (HDD)
- No seek/rotational latency
- No start-up times like HDD

Pollev.com/farhanachoud585

Respond at **PollEv.com/farhanachoud585**

Text **FARHANACHOUD585** to **+61 427 541 357** once to join, then **A or B**

Jane and Anna have 2 identical computers, but Anna's storage device is an SSD while Jane's one is an HDD. When they both switch on their computers at the same time, which one will boot the operating system faster (i.e., loads the OS from disk)?

Jane's HDD **A**

Anna's SSD **B**

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John is buying a new computer for his work. He finalised his choices for all the components except the storage. He cannot decide between a 1TB SSD for $160 (better speed) or a 3TB HDD for $160 (larger capacity). Which one should he choose?

SSD	**A**
HDD	**B**
Isn't it a dilemma we all face?	**C**

What if some more contexts are given?

What does this all mean?

- **Hard disk was the foundation stone for many DBMS design choices**

- Another recent player is networking:
  - More and more databases are distributed
  - The network hardware speeds are close to speed of light already
  - The next determining front for DBMS design choices - some of these are already at play in various systems!

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  - Main memory
  - Type of architecture
  - Communication network
- Software/ DB tuning
  - Types of DB

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Communication Costs

Transmit time = (distance/c) + (message_bits/bandwidth)

c = speed of light (200 million meters/sec) in fibre optics

*This means we can **no longer reduce latency on contemporary hardware further** and increasingly the motto is that the **message length should be large to achieve better utilization**.*

Can you relate with video buffering?

Can you relate the same idea for reading from HDD?

Disk access

Disk access time for HDD

$$Disk \quad access \quad time = seek \quad time + rotational \quad time + \frac{transfer \quad length}{bandwidth}$$

Disk access time for SSD

$$\text{Disk access time} = \frac{transfer \quad length}{bandwidth}$$

Types of Database Systems

Where does spreadsheet fit?

How the data are stored –

Simple file

- As a plain text file. Each line holds one record, with fields separated by delimiters (e.g., commas or tabs)

RDBS

- As a collection of tables (relations) consisting of rows and column. A primary key is used to uniquely identify each row.

Object oriented

- Data stored in the form of ‘objects’ directly (like OOP)

No-SQL

- Non relational – database modelled other than the tabular relations. Covers a wide range of database types.

Types of NoSQL databases

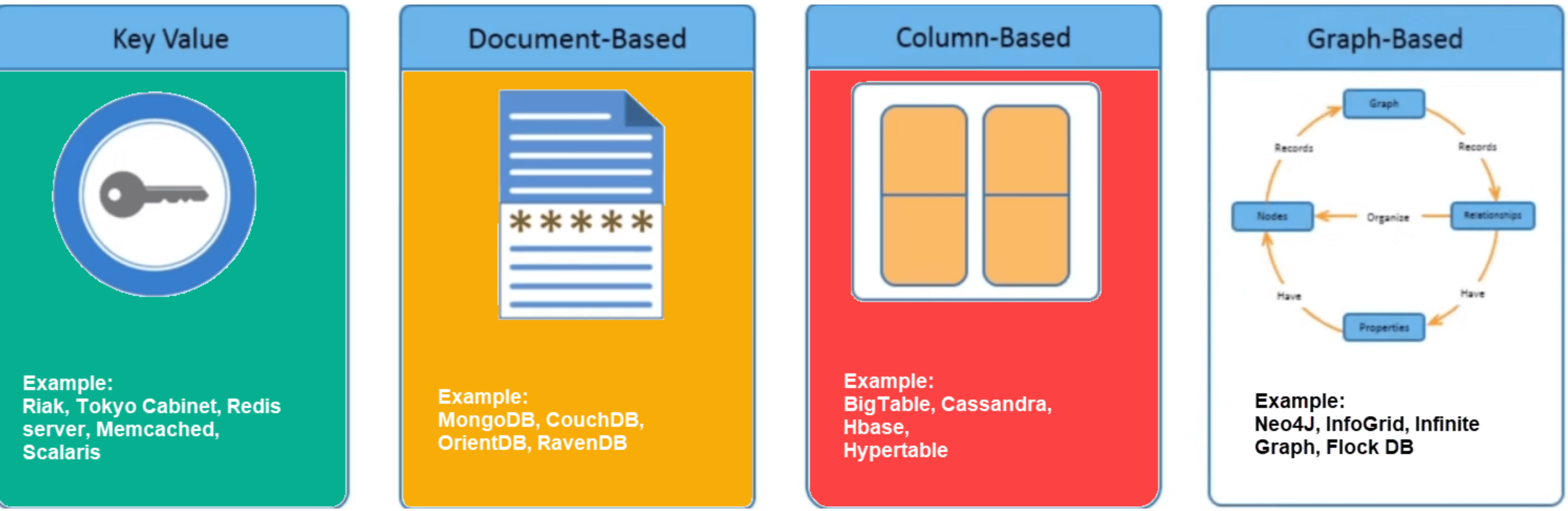

Image source: https://jameskle.com/writes/no-sql

Examples and discussions in tutorials

https://tinyurl.com/24mcctxf

Time for group activities!

Database Architectures

Centralized	Data stored in one location
Distributed	Data distributed across several nodes, can be in different locations
WWW	Stored all over the world, several owners of the data
Grid	Like distributed, but each node manages own resource; system doesn't act as a single unit.
P2P	Like grid, but nodes can join and leave network at will (unlike Grid)
Cloud	Generalization of grid, but resources are accessed on-demand

Each architecture itself is quite a broad topic –
we will only look at what they are and a couple of important properties

Database Architectures

Cloud computing

Updated URL: https://youtu.be/mxT233EdY5c

Summary

- Subject overview
- Hardware components (HDD, SSD)
- Memory hierarchy
- Types of DBs
- DB architecture